

SOLVED PRACTICE WORKBOOK

Emerging Ethics: Technology, AI and Environment - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

Emerging Ethics: Technology, AI and Environment

TEN-SESSION CORE PATH • PRACTICE • OPTIONAL ADVANCED • REGISTER NOTES

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| 1 Emerging ethics: visual foundation and essential vocabulary | 2 How technology and environmental dilemmas arise and are resolved |
| 3 Indian applications, high-yield facts and close-option traps | 4 PYQ entry points, current anchors and Mains theses |
| 5 Digital exclusion, privacy, Puttaswamy and phased DPDP commencement | 6 Selectable evidence units for technology and environmental answers |
| 7 Directive decoding and mark-scaled answer architecture | 8 Environmental orientations, principles and the Vellore boundary |
| 9 Clausewitz routing: separating environmental ethics from war ethics | 10 Study links and complete recent-historical PYQ ownership |

Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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Emerging Ethics: Technology, AI and Environment - Solved Practice Workbook

Generation: learner-v2:g1 **PYQ ownership:** Questions labelled as official reproduce the locally held UPSC GS-IV papers; routed historical demands are explicitly neutral renderings rather than invented quotations.

BASIC MCQS / REMEDIATION

MCQ 1

A welfare department treats a model score as conclusive and refuses to identify any responsible officer. Which principle has been displaced? Which source-grounded ethical principle most precisely explains the case?

- A. Administrative AI can improve speed, consistency and pattern detection, but it should remain a decision-support input whose recommendation is tested against law, context and reasons by an answerable public official.
- B. AI unfairness may originate in unrepresentative data, labels and proxy variables, design objectives, unequal deployment conditions, or feedback loops that turn earlier skewed outputs into future training evidence.
- C. Explainability in high-stakes administration requires an intelligible account of the decisive factors, limits and review route, not merely disclosure of source code or a technical statement that only specialists can understand.
- D. Human oversight is meaningful only when the reviewer has competence, time, authority, relevant information and freedom to depart from an automated recommendation; ceremonial approval preserves neither judgment nor accountability.

Answer: A

Explanation: AI is decision support, not an accountability substitute is the controlling principle. Administrative AI can improve speed, consistency and pattern detection, but it should remain a decision-support input whose recommendation is tested against law, context and reasons by an answerable public official. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 2

A licensing officer checks an AI risk flag against the record and gives independent reasons. Which proper role for AI is illustrated? Which source-grounded ethical principle most precisely explains the case?

- A. Explainability in high-stakes administration requires an intelligible account of the decisive factors, limits and review route, not merely disclosure of source code or a technical statement that only specialists can understand.
- B. Administrative AI can improve speed, consistency and pattern detection, but it should remain a decision-support input whose recommendation is tested against law, context and reasons by an answerable public official.
- C. Human oversight is meaningful only when the reviewer has competence, time, authority, relevant information and freedom to depart from an automated recommendation; ceremonial approval preserves neither judgment nor accountability.
- D. AI unfairness may originate in unrepresentative data, labels and proxy variables, design objectives, unequal deployment conditions, or feedback loops that turn earlier skewed outputs into future training evidence.

Answer: B

Explanation: AI is decision support, not an accountability substitute is the controlling principle. Administrative AI can improve speed, consistency and pattern detection, but it should remain a decision-support input whose recommendation is tested against law, context and reasons by an answerable public official. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 3

A fraud model trained only on urban cases repeatedly misclassifies rural claims. At which bias stage should the diagnosis begin? Which source-grounded ethical principle most precisely explains the case?

- A. Administrative AI can improve speed, consistency and pattern detection, but it should remain a decision-support input whose recommendation is tested against law, context and reasons by an answerable public official.
- B. Explainability in high-stakes administration requires an intelligible account of the decisive factors, limits and review route, not merely disclosure of source code or a technical statement that only specialists can understand.
- C. AI unfairness may originate in unrepresentative data, labels and proxy variables, design objectives, unequal deployment conditions, or feedback loops that turn earlier skewed outputs into future training evidence.
- D. Human oversight is meaningful only when the reviewer has competence, time, authority, relevant information and freedom to depart from an automated recommendation; ceremonial approval preserves neither judgment nor accountability.

Answer: C

Explanation: **Bias can enter data, design, deployment and feedback** is the controlling principle. AI unfairness may originate in unrepresentative data, labels and proxy variables, design objectives, unequal deployment conditions, or feedback loops that turn earlier skewed outputs into future training evidence. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 4

A policing model sends more patrols where it earlier predicted risk, then treats the resulting arrests as proof. Which loop is operating? Which source-grounded ethical principle most precisely explains the case?

- A. Explainability in high-stakes administration requires an intelligible account of the decisive factors, limits and review route, not merely disclosure of source code or a technical statement that only specialists can understand.
- B. Human oversight is meaningful only when the reviewer has competence, time, authority, relevant information and freedom to depart from an automated recommendation; ceremonial approval preserves neither judgment nor accountability.
- C. Administrative AI can improve speed, consistency and pattern detection, but it should remain a decision-support input whose recommendation is tested against law, context and reasons by an answerable public official.
- D. AI unfairness may originate in unrepresentative data, labels and proxy variables, design objectives, unequal deployment conditions, or feedback loops that turn earlier skewed outputs into future training evidence.

Answer: D

Explanation: **Bias can enter data, design, deployment and feedback** is the controlling principle. AI unfairness may originate in unrepresentative data, labels and proxy variables, design objectives, unequal deployment conditions, or feedback loops that turn earlier skewed outputs into future training evidence. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 5

A pension applicant receives only a model version number after rejection. Which ethical safeguard remains unsatisfied? Which source-grounded ethical principle most precisely explains the case?

- A. Explainability in high-stakes administration requires an intelligible account of the decisive factors, limits and review route, not merely disclosure of source code or a technical statement that only specialists can understand.
- B. Administrative AI can improve speed, consistency and pattern detection, but it should remain a decision-support input whose recommendation is tested against law, context and reasons by an answerable public official.
- C. AI unfairness may originate in unrepresentative data, labels and proxy variables, design objectives, unequal deployment conditions, or feedback loops that turn earlier skewed outputs into future training evidence.
- D. Human oversight is meaningful only when the reviewer has competence, time, authority, relevant information and freedom to depart from an automated recommendation; ceremonial approval preserves neither judgment nor accountability.

Answer: A

Explanation: **Explainability must be useful to the affected person** is the controlling principle. Explainability in high-stakes administration requires an intelligible account of the decisive factors, limits and review route, not merely disclosure of source code or a technical statement that only specialists can understand. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 6

An agency states the decisive data, rule, uncertainty and appeal route in plain language. Which concept is being operationalised? Which source-grounded ethical principle most precisely explains the case?

- A. AI unfairness may originate in unrepresentative data, labels and proxy variables, design objectives, unequal deployment conditions, or feedback loops that turn earlier skewed outputs into future training evidence.
- B. Explainability in high-stakes administration requires an intelligible account of the decisive factors, limits and review route, not merely disclosure of source code or a technical statement that only specialists can understand.
- C. Human oversight is meaningful only when the reviewer has competence, time, authority, relevant information and freedom to depart from an automated recommendation; ceremonial approval preserves neither judgment nor accountability.
- D. Administrative AI can improve speed, consistency and pattern detection, but it should remain a decision-support input whose recommendation is tested against law, context and reasons by an answerable public official.

Answer: B

Explanation: **Explainability must be useful to the affected person** is the controlling principle. Explainability in high-stakes administration requires an intelligible account of the decisive factors, limits and review route, not merely disclosure of source code or a technical statement that only specialists can understand. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 7

A junior clerk must approve every model output within ten seconds and cannot alter it. Why is the human-in-the-loop claim hollow? Which source-grounded ethical principle most precisely explains the case?

- A. Administrative AI can improve speed, consistency and pattern detection, but it should remain a decision-support input whose recommendation is tested against law, context and reasons by an answerable public official.
- B. AI unfairness may originate in unrepresentative data, labels and proxy variables, design objectives, unequal deployment conditions, or feedback loops that turn earlier skewed outputs into future training evidence.
- C. Human oversight is meaningful only when the reviewer has competence, time, authority, relevant information and freedom to depart from an automated recommendation; ceremonial approval preserves neither judgment nor accountability.
- D. Explainability in high-stakes administration requires an intelligible account of the decisive factors, limits and review route, not merely disclosure of source code or a technical statement that only specialists can understand.

Answer: C

Explanation: Human oversight must be meaningful is the controlling principle. Human oversight is meaningful only when the reviewer has competence, time, authority, relevant information and freedom to depart from an automated recommendation; ceremonial approval preserves neither judgment nor accountability. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 8

A medical board can pause, investigate and reverse a high-risk model recommendation. Which form of oversight is present? Which source-grounded ethical principle most precisely explains the case?

- A. AI unfairness may originate in unrepresentative data, labels and proxy variables, design objectives, unequal deployment conditions, or feedback loops that turn earlier skewed outputs into future training evidence.
- B. Explainability in high-stakes administration requires an intelligible account of the decisive factors, limits and review route, not merely disclosure of source code or a technical statement that only specialists can understand.
- C. Administrative AI can improve speed, consistency and pattern detection, but it should remain a decision-support input whose recommendation is tested against law, context and reasons by an answerable public official.
- D. Human oversight is meaningful only when the reviewer has competence, time, authority, relevant information and freedom to depart from an automated recommendation; ceremonial approval preserves neither judgment nor accountability.

Answer: D

Explanation: Human oversight must be meaningful is the controlling principle. Human oversight is meaningful only when the reviewer has competence, time, authority, relevant information and freedom to depart from an automated recommendation; ceremonial approval preserves neither judgment nor accountability. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 9

A department blames a vendor after an automated exclusion although no official owned validation. Which governance failure is central? Which source-grounded ethical principle most precisely explains the case?

- A. Algorithmic accountability assigns named responsibility across procurer, developer, deploying department and final decision-maker, while logs, impact assessment, audit trails and incident reporting make conduct reviewable before and after harm.
- B. Contestability means a person can know that automation materially influenced a decision, challenge relevant data and reasoning, obtain timely human reconsideration, and secure correction or remedy without prohibitive cost.
- C. Automation bias is the tendency to over-trust machine output, while de-skilling is the gradual erosion of human expertise through disuse; both require calibrated reliance, training and periodic unaided judgment.
- D. Generative AI may produce fluent but false citations, facts or explanations, so reliability demands source verification, uncertainty disclosure, domain testing and prohibition of unverified output as the sole basis of consequential decisions.

Answer: A

Explanation: Accountability requires an auditable responsibility chain is the controlling principle. Algorithmic accountability assigns named responsibility across procurer, developer, deploying department and final decision-maker, while logs, impact assessment, audit trails and incident reporting make conduct reviewable before and after harm. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 10

A procurement contract preserves logs, assigns incident duties and permits independent audit. Which accountability architecture is strengthened? Which source-grounded ethical principle most precisely explains the case?

- A. Automation bias is the tendency to over-trust machine output, while de-skilling is the gradual erosion of human expertise through disuse; both require calibrated reliance, training and periodic unaided judgment.
- B. Algorithmic accountability assigns named responsibility across procurer, developer, deploying department and final decision-maker, while logs, impact assessment, audit trails and incident reporting make conduct reviewable before and after harm.
- C. Generative AI may produce fluent but false citations, facts or explanations, so reliability demands source verification, uncertainty disclosure, domain testing and prohibition of unverified output as the sole basis of consequential decisions.
- D. Contestability means a person can know that automation materially influenced a decision, challenge relevant data and reasoning, obtain timely human reconsideration, and secure correction or remedy without prohibitive cost.

Answer: B

Explanation: Accountability requires an auditable responsibility chain is the controlling principle. Algorithmic accountability assigns named responsibility across procurer, developer, deploying department and final decision-maker, while logs, impact assessment, audit trails and incident reporting make conduct reviewable before and after harm. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 11

A scholarship applicant can submit corrected records but no one may reconsider the model result. Which element remains absent? Which source-grounded ethical principle most precisely explains the case?

- A. Algorithmic accountability assigns named responsibility across procurer, developer, deploying department and final decision-maker, while logs, impact assessment, audit trails and incident reporting make conduct reviewable before and after harm.
- B. Automation bias is the tendency to over-trust machine output, while de-skilling is the gradual erosion of human expertise through disuse; both require calibrated reliance, training and periodic unaided judgment.
- C. Contestability means a person can know that automation materially influenced a decision, challenge relevant data and reasoning, obtain timely human reconsideration, and secure correction or remedy without prohibitive cost.
- D. Generative AI may produce fluent but false citations, facts or explanations, so reliability demands source verification, uncertainty disclosure, domain testing and prohibition of unverified output as the sole basis of consequential decisions.

Answer: C

Explanation: Contestability gives affected persons an effective remedy is the controlling principle. Contestability means a person can know that automation materially influenced a decision, challenge relevant data and reasoning, obtain timely human reconsideration, and secure correction or remedy without prohibitive cost. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 12

A citizen receives notice, a simple appeal channel and reasoned human review. Which safeguard is demonstrated? Which source-grounded ethical principle most precisely explains the case?

- A. Automation bias is the tendency to over-trust machine output, while de-skilling is the gradual erosion of human expertise through disuse; both require calibrated reliance, training and periodic unaided judgment.
- B. Generative AI may produce fluent but false citations, facts or explanations, so reliability demands source verification, uncertainty disclosure, domain testing and prohibition of unverified output as the sole basis of consequential decisions.
- C. Algorithmic accountability assigns named responsibility across procurer, developer, deploying department and final decision-maker, while logs, impact assessment, audit trails and incident reporting make conduct reviewable before and after harm.
- D. Contestability means a person can know that automation materially influenced a decision, challenge relevant data and reasoning, obtain timely human reconsideration, and secure correction or remedy without prohibitive cost.

Answer: D

Explanation: Contestability gives affected persons an effective remedy is the controlling principle. Contestability means a person can know that automation materially influenced a decision, challenge relevant data and reasoning, obtain timely human reconsideration, and secure correction or remedy without prohibitive cost. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 13

Experienced inspectors stop examining unusual cases because the dashboard is usually right. Which paired risks arise? Which source-grounded ethical principle most precisely explains the case?

- A. Automation bias is the tendency to over-trust machine output, while de-skilling is the gradual erosion of human expertise through disuse; both require calibrated reliance, training and periodic unaided judgment.
- B. Algorithmic accountability assigns named responsibility across procurer, developer, deploying department and final decision-maker, while logs, impact assessment, audit trails and incident reporting make conduct reviewable before and after harm.
- C. Contestability means a person can know that automation materially influenced a decision, challenge relevant data and reasoning, obtain timely human reconsideration, and secure correction or remedy without prohibitive cost.
- D. Generative AI may produce fluent but false citations, facts or explanations, so reliability demands source verification, uncertainty disclosure, domain testing and prohibition of unverified output as the sole basis of consequential decisions.

Answer: A

Explanation: Automation bias and de-skilling are institutional risks is the controlling principle. Automation bias is the tendency to over-trust machine output, while de-skilling is the gradual erosion of human expertise through disuse; both require calibrated reliance, training and periodic unaided judgment. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 14

A department requires periodic manual sampling and records justified departures from model advice. Which risks is it trying to control? Which source-grounded ethical principle most precisely explains the case?

- A. Contestability means a person can know that automation materially influenced a decision, challenge relevant data and reasoning, obtain timely human reconsideration, and secure correction or remedy without prohibitive cost.
- B. Automation bias is the tendency to over-trust machine output, while de-skilling is the gradual erosion of human expertise through disuse; both require calibrated reliance, training and periodic unaided judgment.
- C. Generative AI may produce fluent but false citations, facts or explanations, so reliability demands source verification, uncertainty disclosure, domain testing and prohibition of unverified output as the sole basis of consequential decisions.
- D. Algorithmic accountability assigns named responsibility across procurer, developer, deploying department and final decision-maker, while logs, impact assessment, audit trails and incident reporting make conduct reviewable before and after harm.

Answer: B

Explanation: Automation bias and de-skilling are institutional risks is the controlling principle. Automation bias is the tendency to over-trust machine output, while de-skilling is the gradual erosion of human expertise through disuse; both require calibrated reliance, training and periodic unaided judgment. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 15

An officer files a disciplinary order using model-generated cases without checking them. Which reliability failure is decisive? Which source-grounded ethical principle most precisely explains the case?

- A. Algorithmic accountability assigns named responsibility across procurer, developer, deploying department and final decision-maker, while logs, impact assessment, audit trails and incident reporting make conduct reviewable before and after harm.
- B. Contestability means a person can know that automation materially influenced a decision, challenge relevant data and reasoning, obtain timely human reconsideration, and secure correction or remedy without prohibitive cost.
- C. Generative AI may produce fluent but false citations, facts or explanations, so reliability demands source verification, uncertainty disclosure, domain testing and prohibition of unverified output as the sole basis of consequential decisions.
- D. Automation bias is the tendency to over-trust machine output, while de-skilling is the gradual erosion of human expertise through disuse; both require calibrated reliance, training and periodic unaided judgment.

Answer: C

Explanation: Hallucination and reliability require verification is the controlling principle. Generative AI may produce fluent but false citations, facts or explanations, so reliability demands source verification, uncertainty disclosure, domain testing and prohibition of unverified output as the sole basis of consequential decisions. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 16

A legal cell verifies every generated citation against official records before use. Which control addresses hallucination? Which source-grounded ethical principle most precisely explains the case?

- A. Contestability means a person can know that automation materially influenced a decision, challenge relevant data and reasoning, obtain timely human reconsideration, and secure correction or remedy without prohibitive cost.
- B. Automation bias is the tendency to over-trust machine output, while de-skilling is the gradual erosion of human expertise through disuse; both require calibrated reliance, training and periodic unaided judgment.
- C. Algorithmic accountability assigns named responsibility across procurer, developer, deploying department and final decision-maker, while logs, impact assessment, audit trails and incident reporting make conduct reviewable before and after harm.
- D. Generative AI may produce fluent but false citations, facts or explanations, so reliability demands source verification, uncertainty disclosure, domain testing and prohibition of unverified output as the sole basis of consequential decisions.

Answer: D

Explanation: Hallucination and reliability require verification is the controlling principle. Generative AI may produce fluent but false citations, facts or explanations, so reliability demands source verification, uncertainty disclosure, domain testing and prohibition of unverified output as the sole basis of consequential decisions. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 17

A vendor passes one demonstration and then changes its model without notice. Which procurement controls were missing? Which source-grounded ethical principle most precisely explains the case?

- A. Responsible AI procurement specifies the public purpose, representative testing, security and bias red-teaming, documentation, audit access, change control, incident response, continuing performance monitoring and an exit route.
- B. Deepfakes create an authenticity problem by making fabricated audio or video appear evidentially real; proportionate responses combine provenance signals, verification, rapid correction, platform process and due regard for lawful expression.
- C. Social media dilemmas include falsehood, micro-targeted manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation; speed and reach magnify harm, while blanket censorship can damage expression and democratic debate.
- D. Digital inclusion requires affordable access, devices, language and disability accessibility, literacy, assistance and an effective offline alternative; an online portal can otherwise convert administrative efficiency into substantive exclusion.

Answer: A

Explanation: **Procurement must include red-teaming and monitoring** is the controlling principle. Responsible AI procurement specifies the public purpose, representative testing, security and bias red-teaming, documentation, audit access, change control, incident response, continuing performance monitoring and an exit route. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 18

A department tests adversarial cases before launch and monitors drift after deployment. Which lifecycle approach is illustrated? Which source-grounded ethical principle most precisely explains the case?

- A. Social media dilemmas include falsehood, micro-targeted manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation; speed and reach magnify harm, while blanket censorship can damage expression and democratic debate.
- B. Responsible AI procurement specifies the public purpose, representative testing, security and bias red-teaming, documentation, audit access, change control, incident response, continuing performance monitoring and an exit route.
- C. Digital inclusion requires affordable access, devices, language and disability accessibility, literacy, assistance and an effective offline alternative; an online portal can otherwise convert administrative efficiency into substantive exclusion.
- D. Deepfakes create an authenticity problem by making fabricated audio or video appear evidentially real; proportionate responses combine provenance signals, verification, rapid correction, platform process and due regard for lawful expression.

Answer: B

Explanation: **Procurement must include red-teaming and monitoring** is the controlling principle. Responsible AI procurement specifies the public purpose, representative testing, security and bias red-teaming, documentation, audit access, change control, incident response, continuing performance monitoring and an exit route. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 19

A forged video of a district officer triggers panic before verification. Which ethical value is most directly attacked? Which source-grounded ethical principle most precisely explains the case?

- A. Responsible AI procurement specifies the public purpose, representative testing, security and bias red-teaming, documentation, audit access, change control, incident response, continuing performance monitoring and an exit route.
- B. Social media dilemmas include falsehood, micro-targeted manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation; speed and reach magnify harm, while blanket censorship can damage expression and democratic debate.
- C. Deepfakes create an authenticity problem by making fabricated audio or video appear evidentially real; proportionate responses combine provenance signals, verification, rapid correction, platform process and due regard for lawful expression.
- D. Digital inclusion requires affordable access, devices, language and disability accessibility, literacy, assistance and an effective offline alternative; an online portal can otherwise convert administrative efficiency into substantive exclusion.

Answer: C

Explanation: Deepfakes attack authenticity and public trust is the controlling principle. Deepfakes create an authenticity problem by making fabricated audio or video appear evidentially real; proportionate responses combine provenance signals, verification, rapid correction, platform process and due regard for lawful expression. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 20

Authorities preserve the original, verify provenance and issue a prompt correction without suppressing criticism. Which balanced response is shown? Which source-grounded ethical principle most precisely explains the case?

- A. Social media dilemmas include falsehood, micro-targeted manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation; speed and reach magnify harm, while blanket censorship can damage expression and democratic debate.
- B. Digital inclusion requires affordable access, devices, language and disability accessibility, literacy, assistance and an effective offline alternative; an online portal can otherwise convert administrative efficiency into substantive exclusion.
- C. Responsible AI procurement specifies the public purpose, representative testing, security and bias red-teaming, documentation, audit access, change control, incident response, continuing performance monitoring and an exit route.
- D. Deepfakes create an authenticity problem by making fabricated audio or video appear evidentially real; proportionate responses combine provenance signals, verification, rapid correction, platform process and due regard for lawful expression.

Answer: D

Explanation: Deepfakes attack authenticity and public trust is the controlling principle. Deepfakes create an authenticity problem by making fabricated audio or video appear evidentially real; proportionate responses combine provenance signals, verification, rapid correction, platform process and due regard for lawful expression. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 21

A platform rewards inflammatory falsehood because engagement rises. Which design-linked ethical problem is present? Which source-grounded ethical principle most precisely explains the case?

- A. Social media dilemmas include falsehood, micro-targeted manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation; speed and reach magnify harm, while blanket censorship can damage expression and democratic debate.
- B. Responsible AI procurement specifies the public purpose, representative testing, security and bias red-teaming, documentation, audit access, change control, incident response, continuing performance monitoring and an exit route.
- C. Deepfakes create an authenticity problem by making fabricated audio or video appear evidentially real; proportionate responses combine provenance signals, verification, rapid correction, platform process and due regard for lawful expression.
- D. Digital inclusion requires affordable access, devices, language and disability accessibility, literacy, assistance and an effective offline alternative; an online portal can otherwise convert administrative efficiency into substantive exclusion.

Answer: A

Explanation: Social media combines misinformation, manipulation and privacy harms is the controlling principle. Social media dilemmas include falsehood, micro-targeted manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation; speed and reach magnify harm, while blanket censorship can damage expression and democratic debate. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 22

A public agency counters a rumour with verified facts while protecting children's identities. Which competing values are being balanced? Which source-grounded ethical principle most precisely explains the case?

- A. Deepfakes create an authenticity problem by making fabricated audio or video appear evidentially real; proportionate responses combine provenance signals, verification, rapid correction, platform process and due regard for lawful expression.
- B. Social media dilemmas include falsehood, micro-targeted manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation; speed and reach magnify harm, while blanket censorship can damage expression and democratic debate.
- C. Digital inclusion requires affordable access, devices, language and disability accessibility, literacy, assistance and an effective offline alternative; an online portal can otherwise convert administrative efficiency into substantive exclusion.
- D. Responsible AI procurement specifies the public purpose, representative testing, security and bias red-teaming, documentation, audit access, change control, incident response, continuing performance monitoring and an exit route.

Answer: B

Explanation: Social media combines misinformation, manipulation and privacy harms is the controlling principle. Social media dilemmas include falsehood, micro-targeted manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation; speed and reach magnify harm, while blanket censorship can damage expression and democratic debate. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 23

A village has mobile coverage but elderly pensioners cannot navigate an English-only portal. Which divide persists? Which source-grounded ethical principle most precisely explains the case?

- A. Responsible AI procurement specifies the public purpose, representative testing, security and bias red-teaming, documentation, audit access, change control, incident response, continuing performance monitoring and an exit route.
- B. Deepfakes create an authenticity problem by making fabricated audio or video appear evidentially real; proportionate responses combine provenance signals, verification, rapid correction, platform process and due regard for lawful expression.
- C. Digital inclusion requires affordable access, devices, language and disability accessibility, literacy, assistance and an effective offline alternative; an online portal can otherwise convert administrative efficiency into substantive exclusion.
- D. Social media dilemmas include falsehood, micro-targeted manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation; speed and reach magnify harm, while blanket censorship can damage expression and democratic debate.

Answer: C

Explanation: The digital divide is about capability, not connectivity alone is the controlling principle. Digital inclusion requires affordable access, devices, language and disability accessibility, literacy, assistance and an effective offline alternative; an online portal can otherwise convert administrative efficiency into substantive exclusion. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 24

A service adds assisted access, local-language design and an offline channel. Which ethical principle is advanced? Which source-grounded ethical principle most precisely explains the case?

- A. Deepfakes create an authenticity problem by making fabricated audio or video appear evidentially real; proportionate responses combine provenance signals, verification, rapid correction, platform process and due regard for lawful expression.
- B. Social media dilemmas include falsehood, micro-targeted manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation; speed and reach magnify harm, while blanket censorship can damage expression and democratic debate.
- C. Responsible AI procurement specifies the public purpose, representative testing, security and bias red-teaming, documentation, audit access, change control, incident response, continuing performance monitoring and an exit route.
- D. Digital inclusion requires affordable access, devices, language and disability accessibility, literacy, assistance and an effective offline alternative; an online portal can otherwise convert administrative efficiency into substantive exclusion.

Answer: D

Explanation: The digital divide is about capability, not connectivity alone is the controlling principle. Digital inclusion requires affordable access, devices, language and disability accessibility, literacy, assistance and an effective offline alternative; an online portal can otherwise convert administrative efficiency into substantive exclusion. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 25

A health database is reused for unrelated profiling because the facts were once disclosed voluntarily. Which understanding of privacy rejects this reasoning? Which source-grounded ethical principle most precisely explains the case?

- A. Privacy is not secrecy alone: it protects dignity, decisional autonomy and a person's reasonable control over how information given in one context is collected, combined, inferred and used in another.
- B. A State privacy restriction should be tested for legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards; a useful public objective alone does not complete the inquiry.
- C. G.S.R. 843(E) commenced a specified immediate tranche on 13 November 2025; the Consent Manager tranche begins 13 November 2026, while core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027.
- D. Ethical data processing joins informed and revocable consent with purpose limitation, data minimisation, accuracy, retention limits and reasonable security; a clicked box cannot authorise indefinite collection or unrelated reuse.

Answer: A

Explanation: Privacy protects dignity, autonomy and contextual control is the controlling principle. Privacy is not secrecy alone: it protects dignity, decisional autonomy and a person's reasonable control over how information given in one context is collected, combined, inferred and used in another. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 26

An agency limits reuse to the stated service purpose and permits correction. Which underlying values are protected? Which source-grounded ethical principle most precisely explains the case?

- A. G.S.R. 843(E) commenced a specified immediate tranche on 13 November 2025; the Consent Manager tranche begins 13 November 2026, while core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027.
- B. Privacy is not secrecy alone: it protects dignity, decisional autonomy and a person's reasonable control over how information given in one context is collected, combined, inferred and used in another.
- C. Ethical data processing joins informed and revocable consent with purpose limitation, data minimisation, accuracy, retention limits and reasonable security; a clicked box cannot authorise indefinite collection or unrelated reuse.
- D. A State privacy restriction should be tested for legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards; a useful public objective alone does not complete the inquiry.

Answer: B

Explanation: Privacy protects dignity, autonomy and contextual control is the controlling principle. Privacy is not secrecy alone: it protects dignity, decisional autonomy and a person's reasonable control over how information given in one context is collected, combined, inferred and used in another. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 27

A department cites convenience but identifies no law or narrower alternative before mass tracking. Which constitutional test is incomplete? Which source-grounded ethical principle most precisely explains the case?

- A. Privacy is not secrecy alone: it protects dignity, decisional autonomy and a person's reasonable control over how information given in one context is collected, combined, inferred and used in another.
- B. G.S.R. 843(E) commenced a specified immediate tranche on 13 November 2025; the Consent Manager tranche begins 13 November 2026, while core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027.
- C. A State privacy restriction should be tested for legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards; a useful public objective alone does not complete the inquiry.
- D. Ethical data processing joins informed and revocable consent with purpose limitation, data minimisation, accuracy, retention limits and reasonable security; a clicked box cannot authorise indefinite collection or unrelated reuse.

Answer: C

Explanation: Puttaswamy requires the full proportionality inquiry is the controlling principle. A State privacy restriction should be tested for legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards; a useful public objective alone does not complete the inquiry. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 28

A scheme has legal authority, a legitimate aim, narrow collection, safeguards and review. Which analytical framework supports assessment? Which source-grounded ethical principle most precisely explains the case?

- A. G.S.R. 843(E) commenced a specified immediate tranche on 13 November 2025; the Consent Manager tranche begins 13 November 2026, while core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027.
- B. Ethical data processing joins informed and revocable consent with purpose limitation, data minimisation, accuracy, retention limits and reasonable security; a clicked box cannot authorise indefinite collection or unrelated reuse.
- C. Privacy is not secrecy alone: it protects dignity, decisional autonomy and a person's reasonable control over how information given in one context is collected, combined, inferred and used in another.
- D. A State privacy restriction should be tested for legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards; a useful public objective alone does not complete the inquiry.

Answer: D

Explanation: Puttaswamy requires the full proportionality inquiry is the controlling principle. A State privacy restriction should be tested for legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards; a useful public objective alone does not complete the inquiry. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 29

An answer says every DPDP duty was enforceable from November 2025. Which chronology corrects it? Which source-grounded ethical principle most precisely explains the case?

- A. G.S.R. 843(E) commenced a specified immediate tranche on 13 November 2025; the Consent Manager tranche begins 13 November 2026, while core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027.
- B. Privacy is not secrecy alone: it protects dignity, decisional autonomy and a person's reasonable control over how information given in one context is collected, combined, inferred and used in another.
- C. A State privacy restriction should be tested for legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards; a useful public objective alone does not complete the inquiry.
- D. Ethical data processing joins informed and revocable consent with purpose limitation, data minimisation, accuracy, retention limits and reasonable security; a clicked box cannot authorise indefinite collection or unrelated reuse.

Answer: A

Explanation: DPDP commencement is phased, not fully operational is the controlling principle. G.S.R. 843(E) commenced a specified immediate tranche on 13 November 2025; the Consent Manager tranche begins 13 November 2026, while core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 30

An administrator protects data ethically now while distinguishing future statutory duties. Which law-ethics distinction is shown? Which source-grounded ethical principle most precisely explains the case?

- A. A State privacy restriction should be tested for legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards; a useful public objective alone does not complete the inquiry.
- B. G.S.R. 843(E) commenced a specified immediate tranche on 13 November 2025; the Consent Manager tranche begins 13 November 2026, while core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027.
- C. Ethical data processing joins informed and revocable consent with purpose limitation, data minimisation, accuracy, retention limits and reasonable security; a clicked box cannot authorise indefinite collection or unrelated reuse.
- D. Privacy is not secrecy alone: it protects dignity, decisional autonomy and a person's reasonable control over how information given in one context is collected, combined, inferred and used in another.

Answer: B

Explanation: DPDP commencement is phased, not fully operational is the controlling principle. G.S.R. 843(E) commenced a specified immediate tranche on 13 November 2025; the Consent Manager tranche begins 13 November 2026, while core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 31

A benefits app collects contacts and location unrelated to eligibility under one bundled checkbox. Which safeguards fail? Which source-grounded ethical principle most precisely explains the case?

- A. Privacy is not secrecy alone: it protects dignity, decisional autonomy and a person's reasonable control over how information given in one context is collected, combined, inferred and used in another.
- B. A State privacy restriction should be tested for legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards; a useful public objective alone does not complete the inquiry.
- C. Ethical data processing joins informed and revocable consent with purpose limitation, data minimisation, accuracy, retention limits and reasonable security; a clicked box cannot authorise indefinite collection or unrelated reuse.
- D. G.S.R. 843(E) commenced a specified immediate tranche on 13 November 2025; the Consent Manager tranche begins 13 November 2026, while core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027.

Answer: C

Explanation: Consent must be purpose-bound and security-backed is the controlling principle. Ethical data processing joins informed and revocable consent with purpose limitation, data minimisation, accuracy, retention limits and reasonable security; a clicked box cannot authorise indefinite collection or unrelated reuse. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 32

A service requests only necessary fields, explains use and deletes them when no longer needed. Which data ethic is applied? Which source-grounded ethical principle most precisely explains the case?

- A. A State privacy restriction should be tested for legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards; a useful public objective alone does not complete the inquiry.
- B. G.S.R. 843(E) commenced a specified immediate tranche on 13 November 2025; the Consent Manager tranche begins 13 November 2026, while core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027.
- C. Privacy is not secrecy alone: it protects dignity, decisional autonomy and a person's reasonable control over how information given in one context is collected, combined, inferred and used in another.
- D. Ethical data processing joins informed and revocable consent with purpose limitation, data minimisation, accuracy, retention limits and reasonable security; a clicked box cannot authorise indefinite collection or unrelated reuse.

Answer: D

Explanation: Consent must be purpose-bound and security-backed is the controlling principle. Ethical data processing joins informed and revocable consent with purpose limitation, data minimisation, accuracy, retention limits and reasonable security; a clicked box cannot authorise indefinite collection or unrelated reuse. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 33

A company reports renewable electricity but ignores cooling water and discarded accelerators. Why is its footprint account incomplete? Which source-grounded ethical principle most precisely explains the case?

- A. AI's environmental footprint spans electricity for training and inference, water used in cooling and power generation, embodied impacts of hardware manufacture, mineral extraction, replacement cycles and electronic waste.
- B. Sustainable development meets present needs without compromising future generations' ability to meet their own needs; it neither makes development absolute nor converts environmental protection into an automatic veto.
- C. Intergenerational equity extends public interest across time by requiring present decision-makers to account for durable resource depletion, climate risk and ecological damage imposed on people unable to participate in today's choice.
- D. Rio Principle 15 and Vellore are environmental authorities; their precaution logic may inform high-stakes AI by analogy where uncertain harm could be serious, but they are not binding Indian AI law.

Answer: A

Explanation: AI has energy, water, hardware and e-waste costs is the controlling principle. AI's environmental footprint spans electricity for training and inference, water used in cooling and power generation, embodied impacts of hardware manufacture, mineral extraction, replacement cycles and electronic waste. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 34

A data centre measures operational energy, water, embodied hardware and e-waste. Which lifecycle view is demonstrated? Which source-grounded ethical principle most precisely explains the case?

- A. Intergenerational equity extends public interest across time by requiring present decision-makers to account for durable resource depletion, climate risk and ecological damage imposed on people unable to participate in today's choice.
- B. AI's environmental footprint spans electricity for training and inference, water used in cooling and power generation, embodied impacts of hardware manufacture, mineral extraction, replacement cycles and electronic waste.
- C. Rio Principle 15 and Vellore are environmental authorities; their precaution logic may inform high-stakes AI by analogy where uncertain harm could be serious, but they are not binding Indian AI law.
- D. Sustainable development meets present needs without compromising future generations' ability to meet their own needs; it neither makes development absolute nor converts environmental protection into an automatic veto.

Answer: B

Explanation: AI has energy, water, hardware and e-waste costs is the controlling principle. AI's environmental footprint spans electricity for training and inference, water used in cooling and power generation, embodied impacts of hardware manufacture, mineral extraction, replacement cycles and electronic waste. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 35

A housing plan ignores ecosystem services because present welfare always prevails. Which principle corrects that one-sided claim? Which source-grounded ethical principle most precisely explains the case?

- A. AI's environmental footprint spans electricity for training and inference, water used in cooling and power generation, embodied impacts of hardware manufacture, mineral extraction, replacement cycles and electronic waste.
- B. Intergenerational equity extends public interest across time by requiring present decision-makers to account for durable resource depletion, climate risk and ecological damage imposed on people unable to participate in today's choice.
- C. Sustainable development meets present needs without compromising future generations' ability to meet their own needs; it neither makes development absolute nor converts environmental protection into an automatic veto.
- D. Rio Principle 15 and Vellore are environmental authorities; their precaution logic may inform high-stakes AI by analogy where uncertain harm could be serious, but they are not binding Indian AI law.

Answer: C

Explanation: Sustainable development is a two-generation bridge is the controlling principle. Sustainable development meets present needs without compromising future generations' ability to meet their own needs; it neither makes development absolute nor converts environmental protection into an automatic veto. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 36

A project redesign protects livelihoods while retaining ecological capacity. Which bridging concept supports it? Which source-grounded ethical principle most precisely explains the case?

- A. Intergenerational equity extends public interest across time by requiring present decision-makers to account for durable resource depletion, climate risk and ecological damage imposed on people unable to participate in today's choice.
- B. Rio Principle 15 and Vellore are environmental authorities; their precaution logic may inform high-stakes AI by analogy where uncertain harm could be serious, but they are not binding Indian AI law.
- C. AI's environmental footprint spans electricity for training and inference, water used in cooling and power generation, embodied impacts of hardware manufacture, mineral extraction, replacement cycles and electronic waste.
- D. Sustainable development meets present needs without compromising future generations' ability to meet their own needs; it neither makes development absolute nor converts environmental protection into an automatic veto.

Answer: D

Explanation: Sustainable development is a two-generation bridge is the controlling principle. Sustainable development meets present needs without compromising future generations' ability to meet their own needs; it neither makes development absolute nor converts environmental protection into an automatic veto. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 37

A government discounts irreversible groundwater loss because future residents cannot object. Which ethical duty is neglected? Which source-grounded ethical principle most precisely explains the case?

- A. Intergenerational equity extends public interest across time by requiring present decision-makers to account for durable resource depletion, climate risk and ecological damage imposed on people unable to participate in today's choice.
- B. AI's environmental footprint spans electricity for training and inference, water used in cooling and power generation, embodied impacts of hardware manufacture, mineral extraction, replacement cycles and electronic waste.
- C. Sustainable development meets present needs without compromising future generations' ability to meet their own needs; it neither makes development absolute nor converts environmental protection into an automatic veto.
- D. Rio Principle 15 and Vellore are environmental authorities; their precaution logic may inform high-stakes AI by analogy where uncertain harm could be serious, but they are not binding Indian AI law.

Answer: A

Explanation: **Intergenerational equity represents absent future citizens** is the controlling principle.

Intergenerational equity extends public interest across time by requiring present decision-makers to account for durable resource depletion, climate risk and ecological damage imposed on people unable to participate in today's choice. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 38

A policy applies a long time horizon and preserves ecological options. Which justice principle is operating? Which source-grounded ethical principle most precisely explains the case?

- A. Sustainable development meets present needs without compromising future generations' ability to meet their own needs; it neither makes development absolute nor converts environmental protection into an automatic veto.
- B. Intergenerational equity extends public interest across time by requiring present decision-makers to account for durable resource depletion, climate risk and ecological damage imposed on people unable to participate in today's choice.
- C. Rio Principle 15 and Vellore are environmental authorities; their precaution logic may inform high-stakes AI by analogy where uncertain harm could be serious, but they are not binding Indian AI law.
- D. AI's environmental footprint spans electricity for training and inference, water used in cooling and power generation, embodied impacts of hardware manufacture, mineral extraction, replacement cycles and electronic waste.

Answer: B

Explanation: **Intergenerational equity represents absent future citizens** is the controlling principle.

Intergenerational equity extends public interest across time by requiring present decision-makers to account for durable resource depletion, climate risk and ecological damage imposed on people unable to participate in today's choice. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 39

A note calls Vellore a directly enforceable AI statute. Which doctrinal limit must be restored? Which source-grounded ethical principle most precisely explains the case?

- A. AI's environmental footprint spans electricity for training and inference, water used in cooling and power generation, embodied impacts of hardware manufacture, mineral extraction, replacement cycles and electronic waste.
- B. Sustainable development meets present needs without compromising future generations' ability to meet their own needs; it neither makes development absolute nor converts environmental protection into an automatic veto.
- C. Rio Principle 15 and Vellore are environmental authorities; their precaution logic may inform high-stakes AI by analogy where uncertain harm could be serious, but they are not binding Indian AI law.
- D. Intergenerational equity extends public interest across time by requiring present decision-makers to account for durable resource depletion, climate risk and ecological damage imposed on people unable to participate in today's choice.

Answer: C

Explanation: Precaution may guide high-stakes AI only by analogy is the controlling principle. Rio Principle 15 and Vellore are environmental authorities; their precaution logic may inform high-stakes AI by analogy where uncertain harm could be serious, but they are not binding Indian AI law. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 40

A department pilots and stress-tests an uncertain high-impact AI system before scale-up. Which analogous reasoning supports caution? Which source-grounded ethical principle most precisely explains the case?

- A. Sustainable development meets present needs without compromising future generations' ability to meet their own needs; it neither makes development absolute nor converts environmental protection into an automatic veto.
- B. Intergenerational equity extends public interest across time by requiring present decision-makers to account for durable resource depletion, climate risk and ecological damage imposed on people unable to participate in today's choice.
- C. AI's environmental footprint spans electricity for training and inference, water used in cooling and power generation, embodied impacts of hardware manufacture, mineral extraction, replacement cycles and electronic waste.
- D. Rio Principle 15 and Vellore are environmental authorities; their precaution logic may inform high-stakes AI by analogy where uncertain harm could be serious, but they are not binding Indian AI law.

Answer: D

Explanation: Precaution may guide high-stakes AI only by analogy is the controlling principle. Rio Principle 15 and Vellore are environmental authorities; their precaution logic may inform high-stakes AI by analogy where uncertain harm could be serious, but they are not binding Indian AI law. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 41

A factory pays a fee and claims unlimited discharge is now ethical. Which misunderstanding is present? Which source-grounded ethical principle most precisely explains the case?

- A. Polluter-pays requires the actor causing environmental harm to bear prevention, remediation and compensation costs rather than externalising them to affected communities or the public; it does not purchase permission to pollute.
- B. Environmental justice asks who receives environmental benefits, who bears pollution, displacement and risk, and who has meaningful voice and remedy, with special attention to poor, tribal, nomadic and marginalised communities.
- C. Anthropocentrism values nature chiefly for human interests, biocentrism recognises inherent worth in living beings, and ecocentrism values ecosystems and ecological processes as wholes; administrative reasoning should identify rather than conflate them.
- D. CBDR-RC joins common climate responsibility with historical contribution and capability, while sensitive border projects require security-aware EIA, community consideration, classified handling where necessary, and the least ecologically damaging feasible alternative.

Answer: A

Explanation: Polluter-pays allocates prevention and remediation costs is the controlling principle. Polluter-pays requires the actor causing environmental harm to bear prevention, remediation and compensation costs rather than externalising them to affected communities or the public; it does not purchase permission to pollute. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 42

A regulator requires cleanup, victim compensation and preventive upgrades from the polluter. Which principle is applied? Which source-grounded ethical principle most precisely explains the case?

- A. Anthropocentrism values nature chiefly for human interests, biocentrism recognises inherent worth in living beings, and ecocentrism values ecosystems and ecological processes as wholes; administrative reasoning should identify rather than conflate them.
- B. Polluter-pays requires the actor causing environmental harm to bear prevention, remediation and compensation costs rather than externalising them to affected communities or the public; it does not purchase permission to pollute.
- C. CBDR-RC joins common climate responsibility with historical contribution and capability, while sensitive border projects require security-aware EIA, community consideration, classified handling where necessary, and the least ecologically damaging feasible alternative.
- D. Environmental justice asks who receives environmental benefits, who bears pollution, displacement and risk, and who has meaningful voice and remedy, with special attention to poor, tribal, nomadic and marginalised communities.

Answer: B

Explanation: Polluter-pays allocates prevention and remediation costs is the controlling principle. Polluter-pays requires the actor causing environmental harm to bear prevention, remediation and compensation costs rather than externalising them to affected communities or the public; it does not purchase permission to pollute. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 43

A clean-energy project displaces a tribal community without consultation while distant users gain. Which justice lens reveals the burden? Which source-grounded ethical principle most precisely explains the case?

- A. Polluter-pays requires the actor causing environmental harm to bear prevention, remediation and compensation costs rather than externalising them to affected communities or the public; it does not purchase permission to pollute.
- B. Anthropocentrism values nature chiefly for human interests, biocentrism recognises inherent worth in living beings, and ecocentrism values ecosystems and ecological processes as wholes; administrative reasoning should identify rather than conflate them.
- C. Environmental justice asks who receives environmental benefits, who bears pollution, displacement and risk, and who has meaningful voice and remedy, with special attention to poor, tribal, nomadic and marginalised communities.
- D. CBDR-RC joins common climate responsibility with historical contribution and capability, while sensitive border projects require security-aware EIA, community consideration, classified handling where necessary, and the least ecologically damaging feasible alternative.

Answer: C

Explanation: **Environmental justice tests distribution and voice** is the controlling principle. Environmental justice asks who receives environmental benefits, who bears pollution, displacement and risk, and who has meaningful voice and remedy, with special attention to poor, tribal, nomadic and marginalised communities. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 44

A clearance process includes affected communities, fair compensation and accessible remedy. Which ethical concern is addressed? Which source-grounded ethical principle most precisely explains the case?

- A. Anthropocentrism values nature chiefly for human interests, biocentrism recognises inherent worth in living beings, and ecocentrism values ecosystems and ecological processes as wholes; administrative reasoning should identify rather than conflate them.
- B. CBDR-RC joins common climate responsibility with historical contribution and capability, while sensitive border projects require security-aware EIA, community consideration, classified handling where necessary, and the least ecologically damaging feasible alternative.
- C. Polluter-pays requires the actor causing environmental harm to bear prevention, remediation and compensation costs rather than externalising them to affected communities or the public; it does not purchase permission to pollute.
- D. Environmental justice asks who receives environmental benefits, who bears pollution, displacement and risk, and who has meaningful voice and remedy, with special attention to poor, tribal, nomadic and marginalised communities.

Answer: D

Explanation: **Environmental justice tests distribution and voice** is the controlling principle. Environmental justice asks who receives environmental benefits, who bears pollution, displacement and risk, and who has meaningful voice and remedy, with special attention to poor, tribal, nomadic and marginalised communities. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 45

A wetland is protected solely for drinking-water recharge. Which orientation supplies the stated reason? Which source-grounded ethical principle most precisely explains the case?

- A. Anthropocentrism values nature chiefly for human interests, biocentrism recognises inherent worth in living beings, and ecocentrism values ecosystems and ecological processes as wholes; administrative reasoning should identify rather than conflate them.
- B. Polluter-pays requires the actor causing environmental harm to bear prevention, remediation and compensation costs rather than externalising them to affected communities or the public; it does not purchase permission to pollute.
- C. Environmental justice asks who receives environmental benefits, who bears pollution, displacement and risk, and who has meaningful voice and remedy, with special attention to poor, tribal, nomadic and marginalised communities.
- D. CBDR-RC joins common climate responsibility with historical contribution and capability, while sensitive border projects require security-aware EIA, community consideration, classified handling where necessary, and the least ecologically damaging feasible alternative.

Answer: A

Explanation: Anthropocentric, biocentric and ecocentric reasons differ is the controlling principle.

Anthropocentrism values nature chiefly for human interests, biocentrism recognises inherent worth in living beings, and ecocentrism values ecosystems and ecological processes as wholes; administrative reasoning should identify rather than conflate them. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 46

A river's flow regime is protected as an interdependent ecological whole. Which orientation is most precise? Which source-grounded ethical principle most precisely explains the case?

- A. Environmental justice asks who receives environmental benefits, who bears pollution, displacement and risk, and who has meaningful voice and remedy, with special attention to poor, tribal, nomadic and marginalised communities.
- B. Anthropocentrism values nature chiefly for human interests, biocentrism recognises inherent worth in living beings, and ecocentrism values ecosystems and ecological processes as wholes; administrative reasoning should identify rather than conflate them.
- C. CBDR-RC joins common climate responsibility with historical contribution and capability, while sensitive border projects require security-aware EIA, community consideration, classified handling where necessary, and the least ecologically damaging feasible alternative.
- D. Polluter-pays requires the actor causing environmental harm to bear prevention, remediation and compensation costs rather than externalising them to affected communities or the public; it does not purchase permission to pollute.

Answer: B

Explanation: Anthropocentric, biocentric and ecocentric reasons differ is the controlling principle.

Anthropocentrism values nature chiefly for human interests, biocentrism recognises inherent worth in living beings, and ecocentrism values ecosystems and ecological processes as wholes; administrative reasoning should identify rather than conflate them. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 47

A climate answer says developing states have no duties. Which differentiated principle rejects that conclusion? Which source-grounded ethical principle most precisely explains the case?

- A. Polluter-pays requires the actor causing environmental harm to bear prevention, remediation and compensation costs rather than externalising them to affected communities or the public; it does not purchase permission to pollute.
- B. Environmental justice asks who receives environmental benefits, who bears pollution, displacement and risk, and who has meaningful voice and remedy, with special attention to poor, tribal, nomadic and marginalised communities.
- C. CBDR-RC joins common climate responsibility with historical contribution and capability, while sensitive border projects require security-aware EIA, community consideration, classified handling where necessary, and the least ecologically damaging feasible alternative.
- D. Anthropocentrism values nature chiefly for human interests, biocentrism recognises inherent worth in living beings, and ecocentrism values ecosystems and ecological processes as wholes; administrative reasoning should identify rather than conflate them.

Answer: C

Explanation: CBDR-RC and EIA require differentiated, least-damaging choices is the controlling principle. CBDR-RC joins common climate responsibility with historical contribution and capability, while sensitive border projects require security-aware EIA, community consideration, classified handling where necessary, and the least ecologically damaging feasible alternative. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 48

A border road uses a narrower alignment after rigorous assessment while protecting genuine security information. Which balanced method is shown? Which source-grounded ethical principle most precisely explains the case?

- A. Environmental justice asks who receives environmental benefits, who bears pollution, displacement and risk, and who has meaningful voice and remedy, with special attention to poor, tribal, nomadic and marginalised communities.
- B. Anthropocentrism values nature chiefly for human interests, biocentrism recognises inherent worth in living beings, and ecocentrism values ecosystems and ecological processes as wholes; administrative reasoning should identify rather than conflate them.
- C. Polluter-pays requires the actor causing environmental harm to bear prevention, remediation and compensation costs rather than externalising them to affected communities or the public; it does not purchase permission to pollute.
- D. CBDR-RC joins common climate responsibility with historical contribution and capability, while sensitive border projects require security-aware EIA, community consideration, classified handling where necessary, and the least ecologically damaging feasible alternative.

Answer: D

Explanation: CBDR-RC and EIA require differentiated, least-damaging choices is the controlling principle. CBDR-RC joins common climate responsibility with historical contribution and capability, while sensitive border projects require security-aware EIA, community consideration, classified handling where necessary, and the least ecologically damaging feasible alternative. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

PYQS AND ANSWER PRACTICE

Solved PYQ 1 - 2018 - 20 marks

Question: GS-IV Q10: A big corporate house is engaged in manufacturing industrial chemicals on a large scale. It proposes to set up an additional unit. Many States rejected its proposal due to detrimental effect on the environment. But one State government acceded to the request and permitted the unit close to a city, brushing aside all opposition. The unit was set up 10 years ago and was in full swing till recently. The pollution caused by the industrial effluents was affecting the land, water and crops in the area. It was also causing serious health problems to human beings and animals. This gave rise to a series of agitations demanding the closure of the plant. In a recent agitation thousands of people took part, creating a law and order problem necessitating stern police action. Following the public outcry, the State government ordered the closure of the factory. The closure of the factory resulted in the unemployment of not only those workers who were engaged in the factory but also those who were working in the ancillary units. It also very badly affected those industries which depended on the chemicals manufactured by it. As a senior officer entrusted with the responsibility of handling this issue, how are you going to address it? (Answer in 250 words)

Source / ownership: Exact full case verified against

books\more_previous_papers\GENERAL-STUDIES-PAPER-IV.pdf, page 9. Topic 13 owns pollution, precaution, polluter-pays and environmental-justice substance; Topic 22 supplies the general case-study method.

Model solution

The officer must reject the false choice between uncontrolled production and abrupt livelihood destruction. Stakeholders include polluted communities, workers and their families, ancillary firms, consumers, the company, regulators, police, animals and the local ecosystem. Rights to health, livelihood and peaceful protest compete with public order and economic continuity.

Immediate action should secure drinking water and medical relief, independently sample land, water and effluent, publish non-sensitive findings, and investigate both regulatory failure and any disproportionate policing. Closure should remain effective where serious non-compliance creates continuing harm, but an expert committee should assess whether a time-bound, monitored restart after proven treatment upgrades is environmentally safe. The precautionary principle governs uncertainty; polluter-pays places cleanup, health compensation and restoration costs on the company rather than taxpayers.

Workers need wage support, social-security access, retraining and temporary placement; ancillary firms need transition assistance, not immunity for the polluter. A local grievance forum should include residents, workers, technical experts and administration. Any restart must require zero-bypass monitoring, public compliance reports, escrow for remediation and automatic suspension triggers.

Vellore supports sustainable development, precaution and polluter-pays in environmental law, but the decision remains fact-specific. The ethical settlement therefore restores ecology and health, preserves viable livelihoods where possible, assigns costs to the wrongdoer and replaces reactive agitation with accountable regulation.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 2 - 2020 - 10 marks

Question: GS-IV Q5(b): "The current internet expansion has instilled a different set of cultural values which are often in conflict with traditional values." Discuss. (Answer in 150 words)

Source / ownership: Exact isolated Q5(b) verified against books\more_previous_papers\Gen_St_P4.pdf, page 3. Topic 13 owns the internet-culture subpart; Q5(a), gender inequality and Savitribai Phule, belongs to Topic 2.

Model solution

Internet expansion changes how identity, authority and community are formed. It can promote autonomy, scientific temper, gender voice, intercultural learning and scrutiny of oppressive customs. Young people can find support beyond restrictive local settings, and public institutions can communicate across language and distance.

Conflict arises when network values of speed, visibility, individual self-expression and global consumer culture meet traditions emphasising privacy, hierarchy, family authority and local continuity. The result may be generational tension, imitation, trolling, sexualised harassment, misinformation and polarisation. Yet neither category is morally pure: tradition can preserve solidarity and wisdom but also exclusion; digital culture can liberate but also manipulate attention and commodify intimacy.

The ethical response is critical adaptation, not blanket rejection or surrender. Digital literacy, consent, privacy, respectful dialogue, age-appropriate safeguards and platform accountability should filter harmful practices. Constitutional dignity and equality provide the standard: retain traditions compatible with rights, reform oppressive ones, and domesticate technology through humane civic values.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 3 - 2021 - 10 marks

Question: GS-IV Q2(a): Impact of digital technology as reliable source of input for rational decision making is a debatable issue. Critically evaluate with suitable example. (Answer in 150 words)

Source / ownership: Exact isolated Q2(a) verified against books\more_previous_papers\QP-CSM-21-GENSTUDIESPAPER-IV-110122.pdf, page 2. Topic 13 owns digital decision reliability; Q2(b), innovativeness in ethical dilemmas, is cross-linked to decision method.

Model solution

Digital technology can make decisions more rational by integrating large records, detecting patterns, applying consistent criteria and creating auditable timelines. For example, a district may use satellite and transaction data to prioritise field inspection of suspected benefit leakage, saving scarce staff time.

Reliability, however, is conditional. Incomplete records, biased samples, proxy variables, wrong objectives, cyber manipulation and model drift can produce precise-looking error. A digital score may omit disability, migration or local shocks that an experienced officer would recognise. Automation bias can then convert an advisory flag into an unreasoned denial, while opaque systems prevent citizens from correcting data.

Therefore digital input should be triangulated with verified records and field context. High-stakes use needs data-quality checks, explainable factors, security, human authority to depart, recorded reasons, appeal and periodic audit. Technology is a valuable epistemic aid, not an infallible moral agent. Rationality lies in reviewable evidence and accountable judgment, not in digitisation by itself.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 4 - 2022 - 10 marks

Question: GS-IV Q4(b): Online methodology is being used for day-to-day meetings, institutional approvals in the administration and for teaching and learning in education sector to the extent telemedicine in the health sector is getting popular with the approvals of the competent authority. No doubt, it has advantages and disadvantages for both the beneficiaries and the system at large. Describe and discuss the ethical issues involved in the use of online method particularly to the vulnerable section of the society. (Answer in 150 words)

Source / ownership: Exact isolated Q4(b) verified against books\more_previous_papers\QP-CSM-22-GENERAL-STUDIES-PAPER IV-190922.pdf, page 3. Topic 13 owns online-method ethics; Q4(a), good governance and e-governance initiatives, cross-links to Topic 11.

Model solution

Online methods improve reach, speed, continuity, documentation and reduced travel. Telemedicine can connect specialists to remote patients; online approvals can reduce delay; digital learning can widen access to material.

Their ethical value depends on substantive inclusion. Vulnerable persons may lack devices, connectivity, electricity, literacy, accessible design, private space or local-language support. Women may have less control over household devices; persons with disabilities may face inaccessible interfaces; elderly users may be exposed to fraud. Telemedicine also raises diagnostic limits, informed consent, confidentiality and data-security concerns. An online-only system can transfer administrative cost to the citizen and hide exclusion behind successful portal statistics.

A just design therefore combines assisted digital centres, accessibility standards, plain-language consent, minimal secure data, grievance redress and an effective offline alternative. High-risk health or entitlement decisions need human follow-up. Efficiency is legitimate, but equality, dignity and care require that no person loses an essential service merely because the channel is digital.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 5 - 2023 - 20 marks

Question: GS-IV Q12: You hold a responsible position in a ministry in the government. One day in the morning you received a call from the school of your 11-year-old son that you are required to come and meet the Principal. You proceed to the school and find your son in the Principal's office. The Principal informs you that your son had been found wandering aimlessly in the grounds during the time classes were in progress. The class teacher further informs you that your son has lately become a loner and did not respond to questions in the class; he had also been unable to perform well in the football trials held recently. You bring your son back from the school and in the evening, you along with your wife try to find out the reasons for your son's changed behaviour. After repeated cajoling, your son shares that some children had been making fun of him in the class as well as in the WhatsApp group of the students by calling him stunted, dumb and a frog. He tells you the names of a few children who are the main culprits but pleads with you to let the matter rest. After a few days, during a sporting event, where you and your wife have gone to watch your son play, one of your colleague's son shows you a video in which students have caricatured your son. Further, he also points out the perpetrators who were sitting in the stands. You purposefully walk past them with your son and go home. Next day, you find on social media a video denigrating you, your son and even your wife, stating that you engaged in physical bullying of children on the sports field. The video became viral on social media. Your friends and colleagues began calling you to find out the details. One of your juniors advised you to make a counter video giving the background and explaining that nothing had happened on the field. You, in turn, posted a video which you have captured during the sporting event, identifying the likely perpetrators who were responsible for your son's predicament. You have also narrated what has actually happened in the field and made attempts to bring out the adverse effects of the misuse of social media. (a) Based on the above case study, discuss the ethical issues involved in the use of social media. (b) Discuss the pros and cons of using social media by you to put across the facts to counter the fake propaganda against your family. (Answer in 250 words)

Source / ownership: Exact full case and both demands verified against books\more_previous_papers\QP-CSM-23-GENERAL-STUDIES-PAPER-IV-180923.pdf, pages 14-15. Topic 13 owns social-media ethics; Topic 22 supplies case architecture and child-sensitive implementation.

Model solution

The case involves child dignity and safety, cyberbullying, privacy, misinformation, parental responsibility, public-office restraint, due process and the permanence of digital identification. The original attackers exploited reach and anonymity; the false viral allegation harmed the whole family and public trust. Yet identifying likely minor perpetrators can expose children to retaliation before facts are independently established.

A counter-video offers speed, direct access to the same audience, evidence, correction and resistance to a one-sided narrative. It may reassure colleagues and expose platform misuse. Its disadvantages are escalation, context collapse, further circulation of the child's humiliation, privacy invasion, mistaken identification, vigilantism and the appearance that official status is being used in a private conflict.

I would prioritise the son's psychological safety, counselling and informed participation. I would preserve screenshots and originals, request the school to investigate under a child-protection and anti-bullying process, seek removal or correction through the platform, and use lawful cyber channels if threats or serious defamation persist. Any public response should be brief, factual and anonymised: deny physical bullying, state that evidence has been submitted to competent authorities, and avoid naming minors.

The already posted identifying video should be removed or edited, with a clarification protecting all children. Truthful counter-speech is legitimate, but proportionality and care demand verification, minimum disclosure and institutional remedy rather than a viral contest.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 6 - 2024 - 10 marks

Question: GS-IV Q1(a): The application of Artificial Intelligence as a dependable source of input for administrative rational decision-making is a debatable issue. Critically examine the statement from the ethical point of view. (Answer in 150 words)

Source / ownership: Exact isolated Q1(a) verified against books\mains\05 UPSC 2024 Paper-IV_Final 1.pdf, page 2. Topic 13 is primary owner; Q1(b), dimensions of ethics in professional decision-making, belongs to Topic 1.

Model solution

AI can be a dependable input when it detects patterns across large records, improves consistency, reduces routine delay and enables scarce officials to focus on difficult cases. Used to flag anomalies in welfare payments, it may strengthen stewardship of public funds.

Dependability is not inherent. Bias may enter data, design, deployment and feedback; generative systems may hallucinate; opaque scores weaken reason-giving; automation bias and de-skilling reduce practical wisdom. Accountability also diffuses among vendor, procurer and officer, although the affected citizen still bears the error. From a deontological view, dignity and due process require an intelligible, appealable decision. Consequentialism supports efficiency only after error and distributional harms are counted.

High-stakes administrative AI should therefore remain advisory. Representative testing, red-teaming, audit logs, plain-language explanations, data correction, meaningful human review, contestability and continuous monitoring are essential. The India AI Governance Guidelines provide policy guidance, not proof of fairness. AI is ethically dependable only as a verified and supervised input within an accountable human institution.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 7 - 2024 - 10 marks

Question: GS-IV Q2(b): Global warming and climate change are the outcomes of human greed in the name of development, indicating the direction in which extinction of organisms including human beings is heading towards loss of life on Earth. How do you put an end to this to protect life and bring equilibrium between the society and the environment? (Answer in 150 words)

Source / ownership: Exact isolated Q2(b) verified against books\mains\05 UPSC 2024 Paper-IV_Final 1.pdf, page 2. Topic 13 owns climate ethics; Q2(a), powerful nations and ongoing conflicts, belongs to Topic 12.

Model solution

The statement identifies greed as a vice: limitless consumption treats nature, poorer communities and future generations as instruments. Climate change is therefore not only a technical market failure but a failure of temperance, justice and stewardship. Its burdens are unequal: those contributing least often face the greatest heat, livelihood and disaster risks.

Equilibrium requires sustainable development: efficient and renewable energy, public transport, circular production, ecosystem restoration, climate-resilient agriculture and credible disclosure by firms. Polluter-pays should internalise damage, while precaution supports action despite uncertainty about exact local effects. Citizens must moderate wasteful consumption, but responsibility cannot be shifted from institutions to individual lifestyle alone.

Internationally, CBDR-RC combines common action with historical contribution and respective capability; it means differentiated, not absent, obligations for developing states. Participatory transitions and support for affected workers make mitigation just. The goal is not to end development, but to redefine it so present welfare no longer destroys the ecological conditions of future freedom and life.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 8 - 2024 - 20 marks

Question: GS-IV Q7: There is a technological company named ABC Incorporated which is the second largest worldwide, situated in the Third World. You are the Chief Executive Officer and the majority shareholder of this company. The fast technological improvements have raised worries among environmental activists, regulatory authorities, and the general public over the sustainability of this scenario. You confront substantial issues about the business's environmental footprint. In 2023, your organization had a significant increase of 48% in greenhouse gas emissions compared to the levels recorded in 2019. The significant rise in energy consumption is mainly due to the surging energy requirements of your data centers, fuelled by the exponential expansion of Artificial Intelligence (AI). AI-powered services need much more computational resources and electrical energy compared to conventional online activities, notwithstanding their notable gains. The technology's proliferation has led to a growing concern over the environmental repercussions, resulting in an increase in warnings. AI models, especially those used in extensive machine learning and data processing, exhibit much greater energy consumption than conventional computer tasks, with an exponential increase. Although there is already a commitment and goal to achieve net zero emissions by 2030, the challenge of lowering emissions seems overwhelming as the integration of AI continues to increase. To achieve this goal, substantial investments in renewable energy use would be necessary. The difficulty is exacerbated by the competitive environment of the technology sector, where rapid innovation is essential for preserving market standing and shareholders' worth. To achieve a balance between innovation, profitability and sustainability, a strategic move is necessary that is in line with both business objectives and ethical obligations. (a) What is your immediate response to the challenges posed in the above case? (b) Discuss the ethical issues involved in the above case. (c) Your company has been identified to be penalized by technological giants. What logical and ethical arguments will you put forth to convince about its necessity? (d) Being a conscience being, what measures would you adopt to maintain balance between AI innovation and environmental footprint? (Answer in 250 words)

Source / ownership: Exact full case and all four demands verified against books\mains\05 UPSC 2024 Paper-IV_Final 1.pdf, pages 4-5. Topic 13 owns AI-footprint and environmental substance; Topic 22 owns method and Topic 12 is a corporate-governance cross-link. The official case contains no algorithmic-discrimination fact.

Model solution

I would acknowledge the 48% increase, freeze avoidable high-carbon expansion pending a rapid independent footprint review, and protect essential services while the board adopts a verified transition plan. Stakeholders are users, employees, shareholders, regulators, energy and water-stressed communities, future generations and competitors.

The ethical issues are truthful disclosure versus greenwashing; profit and innovation versus climate duty; present shareholder value versus intergenerational equity; unequal burdens in a Third World setting; and responsibility for operational energy, cooling water, hardware and e-waste. A 2030 promise cannot excuse present inaction. Polluter-pays and sustainable development require costs to be internalised rather than shifted.

If stronger technological giants seek to penalise ABC, I would argue for transparent, uniform and capability-sensitive standards, independent verification and a fair transition period. Environmental accountability is necessary, but private rivals should not use it as discriminatory market exclusion. Equivalent footprints must face equivalent rules, with CBDR-RC used as an ethical analogy for capability, not as corporate immunity.

Measures include science-based interim carbon budgets; renewable procurement plus additional clean capacity; efficient models, hardware utilisation and workload scheduling; water-aware siting and cooling; repair, reuse and responsible e-waste; supplier standards; internal carbon pricing; audited public reporting; and executive incentives tied to absolute reductions. A board sustainability committee should monitor milestones and publish setbacks. Innovation should continue within resource budgets, with low-value computation deferred and high-social-value uses prioritised.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 9 - 2024 - 20 marks

Question: GS-IV Q12: Dr. Srinivasan is a senior scientist working for a reputed biotechnology company known for its cutting-edge research in pharmaceuticals. Dr. Srinivasan is heading a research team working on a new drug aimed at treating a rapidly spreading variant of a new viral infectious disease. The disease has been rapidly spreading across the world and the cases reported in the country are increasing. There is huge pressure on Dr. Srinivasan's team to expedite the trials for the drug as there is significant market for it, and the company wants to get the first-mover advantage in the market. During a team meeting, some senior team members suggest some shortcut for expediting the clinical trials for the drug and for getting the requisite approvals. These include manipulating data to exclude some negative outcomes and selectively reporting positive results, foregoing the process of informed consent and using compounds already patented by a rival company, rather than developing one's own component. Dr. Srinivasan is not comfortable taking such shortcuts, at the same time he realises meeting the targets is impossible without using these means. (a) What would you do in such a situation? (b) Examine your options and consequences in the light of the ethical questions involved. (c) How can data ethics and drug ethics save humanity at large in such a scenario? (Answer in 250 words)

Source / ownership: Exact full case and all three demands verified against books\mains\05 UPSC 2024 Paper-IV_Final 1.pdf, page 10. Topic 13 owns the data-integrity and consent cross-link; Topic 22 owns case method, with research and drug ethics supplying the specialist substantive route.

Model solution

I would refuse data manipulation, selective reporting, consent bypass and unauthorised use of patented compounds; preserve records; document the pressure; seek an urgent independent ethics and regulatory review; and propose a lawful accelerated protocol. Patients, trial participants, future users, researchers, the company, the rival patent holder, regulators and the wider public are stakeholders.

Compliance with the shortcuts may produce speed and first-mover gain, but it converts participants into means, conceals adverse effects, corrupts scientific knowledge, risks mass injury and destroys trust. Silent refusal protects personal integrity but leaves organisational pressure intact. Resignation may be necessary if correction fails, yet premature exit can abandon evidence. The strongest option is recorded internal escalation, independent review and competent external reporting where serious misconduct persists, with protection of confidential patient data.

Data ethics requires completeness, accuracy, reproducibility, secure handling, transparent adverse-result reporting and correction. Drug ethics requires scientific validity, informed and voluntary consent, favourable risk-benefit

assessment, fair participant selection, monitoring and post-trial responsibility. Patent rights must be respected or addressed through lawful licensing and applicable public-health mechanisms, not covert appropriation.

A lawful accelerated design may use adaptive protocols, parallel administrative preparation and rapid independent review without weakening consent or evidence. These disciplines save humanity because unreliable data in a fast-moving epidemic scales harm globally. Urgency justifies faster ethical work, not lower truth, dignity or safety.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 10 - 2025 - 10 marks

Question: GS-IV Q1(a): In the present digital age, social media has revolutionised our way of communication and interaction. However, it has raised several ethical issues and challenges. Describe the key ethical dilemmas in this regard. (Answer in 150 words)

Source / ownership: Exact isolated Q1(a) verified against books\mains\UPSC Mains 2025 GS Paper 4.pdf, page 2. Topic 13 owns social-media ethics; Q1(b), constitutional morality and public service, belongs to Topic 14.

Model solution

Social media expands voice, association, emergency communication and accountability, but its ethical dilemmas arise from a business and design environment that rewards attention. Freedom of expression conflicts with protection from incitement, cyberbullying and defamation. Personalisation offers relevance but enables surveillance, micro-targeted manipulation and loss of autonomy. Virality democratises information yet accelerates misinformation, deepfakes and irreversible reputational harm before verification.

Further dilemmas include privacy versus public curiosity, anonymity versus answerability, platform moderation versus censorship, child safety versus participation, and global rules versus linguistic and cultural context. Algorithmic amplification can create polarisation and silence minorities even without direct state coercion.

An ethical response combines user digital literacy, consent and privacy by design, child-sensitive defaults, provenance and rapid correction tools, transparent moderation reasons, appeal, researcher access and proportionate lawful regulation. Government counter-speech should be factual and restrained. The aim is not a sanitised public sphere, but one where expression remains free while manipulation, hidden surveillance and preventable harm become contestable and accountable.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 11 - 2025 - 10 marks

Question: GS-IV Q2(b): Keeping the national security in mind, examine the ethical dilemmas related to controversies over environmental clearance of development projects in ecologically sensitive border areas in the country. (Answer in 150 words)

Source / ownership: Exact isolated Q2(b) verified against books\mains\UPSC Mains 2025 GS Paper 4.pdf, page 2. Topic 13 is the sole substantive owner of this subpart; Q2(a), Clausewitz and war as diplomacy, belongs to Topic 12.

Model solution

Border roads, communications and logistics can protect sovereignty, responders and remote communities. Yet ecologically sensitive mountains and forests may face irreversible habitat fragmentation, erosion, disaster risk and livelihood loss. The dilemma is therefore security and development versus precaution, community voice and intergenerational equity; neither value may be reduced to a slogan.

Blanket exemption risks using secrecy to avoid scrutiny. Complete public disclosure, however, may reveal genuine operational information. A proportionate process should define the specific security purpose, compare routes and technologies, choose the least ecologically damaging feasible alternative, consult affected communities on non-classified impacts, and subject classified material to cleared independent experts. Cumulative and disaster-risk assessment, seasonal construction controls, compensatory restoration, wildlife passages and continuous monitoring should follow.

Precaution does not ban all border infrastructure; it demands safeguards despite uncertainty. Reasons, conditions and compliance results should be public except for narrowly protected security details. Ethical clearance is thus a reviewable, least-damage accommodation of a legitimate security need, not automatic approval or automatic prohibition.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 12 - 2025 - 20 marks

Question: GS-IV Q8: In line with the Directive Principles of State Policy enshrined in the Indian Constitution, the government has a constitutional obligation to ensure basic needs - "Roti, Kapda aur Makan (Food, Clothes and Shelter)" - for the under-privileged. Pursuing this mandate, the district administration proposed clearing a portion of forest land to develop housing for the homeless and economically weaker sections of the society. The proposed land, however, is an ecologically sensitive zone densely populated with age-old trees, medicinal plants and vital biodiversity. Besides, these forests help to regulate micro-climate and rainfalls, provide habitat for wildlife, support soil fertility and prevent land/soil erosion and sustain livelihoods of tribal and nomadic communities. In spite of the ecological and social costs, the administration argues in favour of the said proposal by highlighting that this very initiative addresses fundamental human rights as a critical welfare priority. Besides, it fulfils the government's duty to uplift and empower the poor through inclusive housing development. Further, these forest areas have become unsafe due to wild-animal threats and recurring human-wild life conflicts. Lastly, clearing forest-zones may help to curb anti-social elements allegedly using these areas as hideouts, thereby enhancing law and order. (a) Can deforestation be ethically justified in the pursuit of social welfare objectives like, housing for the homeless? (b) What are the socio-economic, administrative and ethical challenges in balancing environmental conservation with human development? (c) What substantial alternatives or policy interventions can be proposed to ensure that both environmental integrity and human dignity are protected? (Answer in 250 words)

Source / ownership: Exact full case and all three demands verified against books\mains\UPSC Mains 2025 GS Paper 4.pdf, pages 5-6. Topic 13 owns environmental substance; Topic 22 owns method. No degraded-forest, city-periphery or court-monitoring fact is added.

Model solution

Housing is a dignity and welfare imperative, but the stated forest is not vacant land. It contains age-old trees, medicinal plants, biodiversity, climate and rainfall regulation, wildlife habitat, fertile soil, erosion control and tribal and nomadic livelihoods. These ecological services support human rights across generations. Deforestation is therefore justifiable only after necessity and least-damaging alternatives are proved, not merely because the beneficiary goal is worthy.

Socio-economic challenges include land scarcity, affordability, livelihood displacement and human-wildlife conflict. Administrative challenges include reliable beneficiary selection, interdepartmental coordination, ecological valuation, consultation, rehabilitation and preventing capture. Ethical tensions oppose immediate shelter to intergenerational equity, anthropocentric welfare to biocentric and ecocentric value, and majority benefit to the voice of tribal and nomadic communities. Alleged hideouts do not establish that clearing is a necessary law-and-order response.

The administration should first inventory public land, vacant serviced plots and brownfield sites; use in-situ upgrading, rental housing, land pooling, vertical construction and distributed smaller sites; and improve transport to suitable land. It should map ecosystem services and wildlife corridors, consult affected communities, and compare lifecycle costs. If a residual diversion is demonstrably unavoidable, minimise its footprint, protect critical habitat and medicinal species, secure lawful livelihood rehabilitation, create wildlife mitigation and fund long-term restoration and monitoring.

The ethical verdict is integrated planning: fulfil shelter without treating the poor as a pretext for avoidable ecological loss or conservation as a reason to deny human dignity.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

ORIGINAL MAINS PRACTICE WITH MODEL SOLUTIONS

Original Mains Practice 1 - 10 marks

Question: Administrative AI should support judgment without becoming an unanswerable final decision-maker. Examine with suitable safeguards. Answer in about 150 words.

Model solution

Administrative AI can improve speed, consistency, anomaly detection and allocation of scarce inspection capacity. A welfare department may use it to flag unusual claims for human review. However, machine output reflects chosen data, labels, objectives and deployment conditions. It may reproduce exclusion, hallucinate, drift or omit context. Automation bias then makes a recommendation appear objective, while accountability is diffused among vendor, procurer and official.

The ethical position is decision support under human responsibility. Before procurement, the department should define purpose, rights impact and success criteria. It should require representative testing, security and bias red-teaming, documentation, audit access and an exit route. Deployment needs meaningful human authority to depart, intelligible reasons, notice of automation, correction of data, appeal and continuous monitoring. High-stakes decisions should never rest solely on unverified generative output.

The India AI Governance Guidelines offer useful policy guidance but do not prove a system fair or safe. AI becomes rational only inside a reviewable institution where an identified human remains answerable.

Why this earns marks: The response defines the issue, develops the relevant mechanism with India-centric evidence, tests a limitation, and gives a mark-scaled conclusion.

Original Mains Practice 2 - 10 marks

Question: Why are explainability, contestability and accountability distinct but mutually dependent safeguards in high-stakes AI? Answer in about 150 words.

Model solution

Explainability supplies intelligible reasons: the affected person should know the decisive factors, important limits and role of automation. Contestability converts knowledge into agency by allowing data correction, challenge, timely human reconsideration and remedy. Accountability assigns institutional responsibility and preserves logs, audits and sanctions so that failure has an answerable owner.

Each is insufficient alone. A technically accurate explanation without appeal merely describes an injustice. An appeal without reasons forces the citizen to guess what to challenge. Reasons and appeal without responsibility may end in the vendor and department blaming each other. In a scholarship-rejection system, the applicant should receive the relevant eligibility factors, correct an erroneous record and obtain a reasoned review by an authorised officer; the agency should retain audit logs and investigate recurring error.

Commercial secrecy may protect genuine intellectual property, but cannot erase due process. Risk-sensitive disclosure, independent confidential audit and plain-language adverse-action reasons can reconcile both interests. Together the three safeguards make automated power understandable, challengeable and answerable.

Why this earns marks: The response defines the issue, develops the relevant mechanism with India-centric evidence, tests a limitation, and gives a mark-scaled conclusion.

Original Mains Practice 3 - 15 marks

Question: A consent form alone does not make administrative data use ethical. Discuss through privacy, purpose limitation and the phased DPDP framework. Answer in about 200 words.

Model solution

Consent protects autonomy only when it is informed, specific, accessible and capable of withdrawal. In public administration, dependence on an essential service can make formal agreement weak: a citizen may click a bundled notice because refusal means losing a benefit. Ethical data use therefore also requires lawful authority, purpose limitation, minimisation, accuracy, retention limits, security and remedy.

Puttaswamy recognises privacy as a fundamental right linked to dignity and autonomy. A State restriction requires legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards. A useful scheme objective alone is insufficient. Thus a health database collected for treatment should not silently become a general profiling system.

The DPDP framework must be stated chronologically. G.S.R. 843(E) commenced specified provisions on 13 November 2025; the Consent Manager tranche begins 13 November 2026; core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027. It is wrong to call the Act fully operational at the present cut-off.

Nevertheless, deferred statutory commencement does not defer ethics. Administrators should use plain-language and granular notices, collect only necessary data, separate optional reuse, secure records, publish retention rules and provide correction and grievance channels. Consent is one part of a dignity-preserving architecture, not a universal moral waiver.

Why this earns marks: The response defines the issue, develops the relevant mechanism with India-centric evidence, tests a limitation, and gives a mark-scaled conclusion.

Original Mains Practice 4 - 15 marks

Question: Can the environmental precautionary principle guide governance of high-stakes AI? Critically examine the analogy and its limits. Answer in about 200 words.

Model solution

Precaution addresses decisions where scientific uncertainty coexists with threats of serious or irreversible harm. Rio Declaration Principle 15 gives its classic environmental formulation, and Vellore recognises precaution within Indian environmental law. High-stakes AI presents a structural analogy: model error may be opaque, scale rapidly and burden people who lack voice, while evidence of harm emerges only after deployment.

The analogy supports staged pilots, impact assessment, adversarial red-teaming, security tests, representative evaluation, fallback procedures, human oversight and continuous monitoring before mass use in welfare, policing or health. Lack of perfect certainty should not justify exposing citizens to an untested irreversible denial.

However, Rio and Vellore are environmental authorities, not binding Indian AI law. AI harms may be reversible through correction, while blanket precaution can freeze socially beneficial innovation and entrench incumbent vendors. Risk also varies: a translation assistant and an automated liberty decision cannot face identical controls.

Therefore precaution should operate as a proportionate ethical analogy. Classify impact, require stronger evidence where rights and irreversibility are greater, preserve contestability and review controls as performance changes. The principle justifies anticipatory governance, not a general technology veto or a false claim that Vellore directly regulates AI.

Why this earns marks: The response defines the issue, develops the relevant mechanism with India-centric evidence, tests a limitation, and gives a mark-scaled conclusion.

Original Mains Practice 5 - 20 marks

Question: Examine the full environmental footprint of AI through sustainable development and intergenerational justice. Suggest an accountable transition framework for India. Answer in about 250 words.

Model solution

AI's footprint is a lifecycle issue, not electricity alone. Training and repeated inference consume power; cooling and power generation consume water; semiconductor and server manufacture embody energy, minerals and pollution; rapid replacement produces electronic waste. Benefits and burdens are also spatially unequal: urban and global users may gain while water-stressed communities, workers near extraction and future generations bear costs.

Sustainable development requires meeting present needs without compromising future capacity. Intergenerational justice gives absent future citizens standing in today's resource allocation. Polluter-pays opposes shifting remediation to taxpayers, and environmental justice requires voice for affected communities. Yet a blanket restriction would ignore AI's public value in health, agriculture, disaster warning and accessible services.

An accountable Indian transition should begin with standardised disclosure of operational energy, water, location, embodied hardware and e-waste, without unsupported precision where measurement is immature. Public procurement can reward efficient models, repairability, renewable additionality, water-aware siting and responsible end-of-life handling. Operators should use carbon and water budgets, efficient architectures, high utilisation, workload scheduling, heat reuse where feasible and verified supplier standards. EIA and local consultation should apply where the governing law requires them; environmental claims need independent assurance.

Policy should prioritise high-social-value computation, support clean grids and domestic circular hardware capacity, and protect workers and communities during transition. Periodic public review must compare absolute impacts, not efficiency claims alone. The ethical goal is useful AI within ecological budgets, not innovation whose hidden costs narrow future freedom.

Why this earns marks: The response defines the issue, develops the relevant mechanism with India-centric evidence, tests a limitation, and gives a mark-scaled conclusion.

Original Mains Practice 6 - 20 marks

Question: A State proposes an AI system to select welfare beneficiaries and a water-intensive data centre to operate it in a drought-prone district. As district head, resolve the combined data, justice and environmental dilemma. Answer in about 250 words.

Model solution

Stakeholders include eligible applicants, excluded families, local residents and farmers, tribal or marginalised groups, officials, the vendor, data-centre workers, the power and water utilities, the ecosystem and future citizens. The proposal promises faster targeting and jobs, but creates linked risks: biased exclusion, opaque reasons, privacy loss, vendor dependence, groundwater stress, energy emissions and unequal distribution of benefits and burdens.

I would not approve immediate full-scale deployment. First, separate and test the two decisions. For beneficiary selection, define lawful eligibility, prohibit the model from making final adverse decisions, audit data and proxies, red-team vulnerable cases, provide notice, plain-language reasons, data correction, assisted offline application and timely human appeal. Contracts must assign accountability, preserve logs, permit independent audit and allow exit. Puttaswamy proportionality and purpose limitation govern personal-data use.

For the data centre, commission a transparent assessment of water source, drought scenarios, energy, cooling, hardware and e-waste; compare less water-intensive locations and architectures; consult affected communities; and select the least-damaging feasible alternative. Approval, if justified, should impose water and carbon budgets, renewable additionality, non-potable or recycled water where safe, public monitoring, drought curtailment triggers and restoration security. Claimed welfare benefits cannot override ecological limits.

A limited pilot using existing lower-impact infrastructure may proceed if independent review confirms rights and reliability safeguards. The final decision should publish reasons and residual risks. Innovation is acceptable only when beneficiaries can contest digital power and host communities do not subsidise it with dignity, water and future resilience.

Why this earns marks: The response defines the issue, develops the relevant mechanism with India-centric evidence, tests a limitation, and gives a mark-scaled conclusion.